

From 2 AM Failure to 8 AM Fix: Self-Healing Docs, Agent Instructions Included

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YOUR SPEAKER

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15 YEARS	Building documentation for Apple, Google, and Skyflow
OPEN SOURCE	Created Doc Detective, a documentation testing tool
RESEARCH	Co-authored DoGBench, the Documentation Generation Benchmark
BOOKS	*Docs as Tests*, *Docs as Tests & AI*, and the forthcoming *Docs as Tests & Hardware*

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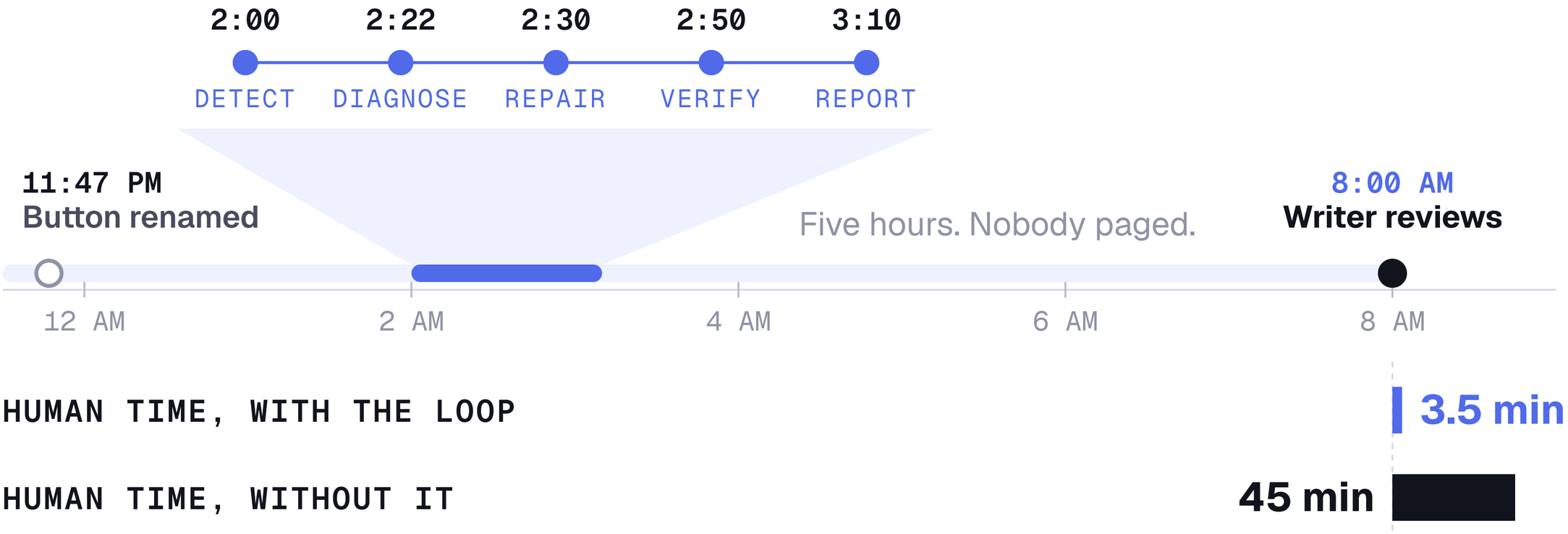
11:47 PM

A release renames a button. Nobody tells the docs team.

PR #4521 merges. `Login` becomes `Sign In`. Three pages and one screenshot still say `Login`.

THE NIGHT SHIFT

By 8 AM, the fix is waiting for review



Illustrative scenario adapted from *Docs as Tests & AI*, ch. 18

THE DIFFERENCE

The writer spent three and a half minutes. It used to take 45.

3.5 min

With the loop. Read the evidence, check the diff, approve.

45 min

Without it. Trace the change, fix three pages, re-run the tests.

WHERE WE'RE GOING

You'll leave with a loop, a ladder, and a scoreboard

The loop

Five steps, four failures

The ladder

Five levels, and where to stop

One more doc set

Agent instructions drift too

The scoreboard

Metrics, and when to step back

Build and bound

A one-day build, and its limits

THE FIVE STEPS

Each step earns its place by what it hands to the next



DETECT

Diagnosis is only as good as the test description

THE AGENT GETS

```
Test 47  FAILED
```

Starts from nothing

THE AGENT NEEDS

```
Login button is present
on getting-started page
FAILED

"Login" not found
```

Half the diagnosis is done

DIAGNOSE

Diagnosis turns "test failed" into a cause

Root cause	PR #4521
	"Login" → "Sign In"
Scope	3 pages
	1 screenshot
Confidence	High

THE AGENT READS

- Change history
- Release notes and deploy logs
- Every page that mentions the change

REPAIR AND VERIFY

Repair narrowly, then prove it before anyone looks

REPAIR

```
- Login
+ Sign In
```

3 pages, 1 screenshot, then stop

VERIFY

- Failing test passes
- Full suite passes
- Screenshot matches

REPORT

A self-healing pull request carries its own evidence

```
### Self-Healing Fix: UI string update

Root cause:    PR #4521 (UI cleanup) renamed
               button "Login" → "Sign In"
Trigger:       Nightly test failure: button
               with text "Login" not found
Changes:       3 pages
  getting-started.md   2 instances
  authentication.md    1 instance
  troubleshooting.md   caption, screenshot
Verification: 47 tests passing, 0 regressions
Limitations:   None. String swap only.
Attribution:   Generated by self-healing system
```

Summary	One line: what changed and why
Root cause	The product change that triggered it
Evidence	Failing run, product PR, correlation
Changes	The diff, scoped narrowly
Verification	What passed, what warned
Limitations	What it couldn't fix, and why
Attribution	A clear marker that a system wrote it

FOUR FAILURES YOU ACTUALLY HAVE

Same loop, different middle

API changes

CAUGHT BY

Contract test

SHIPS AS

PR, extra review if breaking

Broken links

CAUGHT BY

Link checker

SHIPS AS

PR or issue

Stale screenshots

CAUGHT BY

Visual diff

SHIPS AS

PR with before and after

Conceptual drift

CAUGHT BY

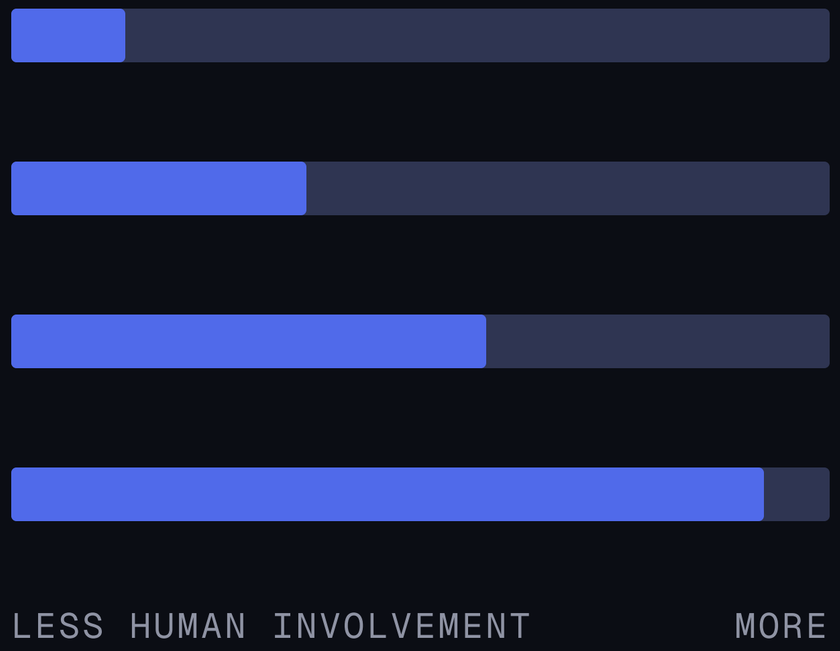
Release notes

SHIPS AS

Draft for a person

WHERE PEOPLE COME IN

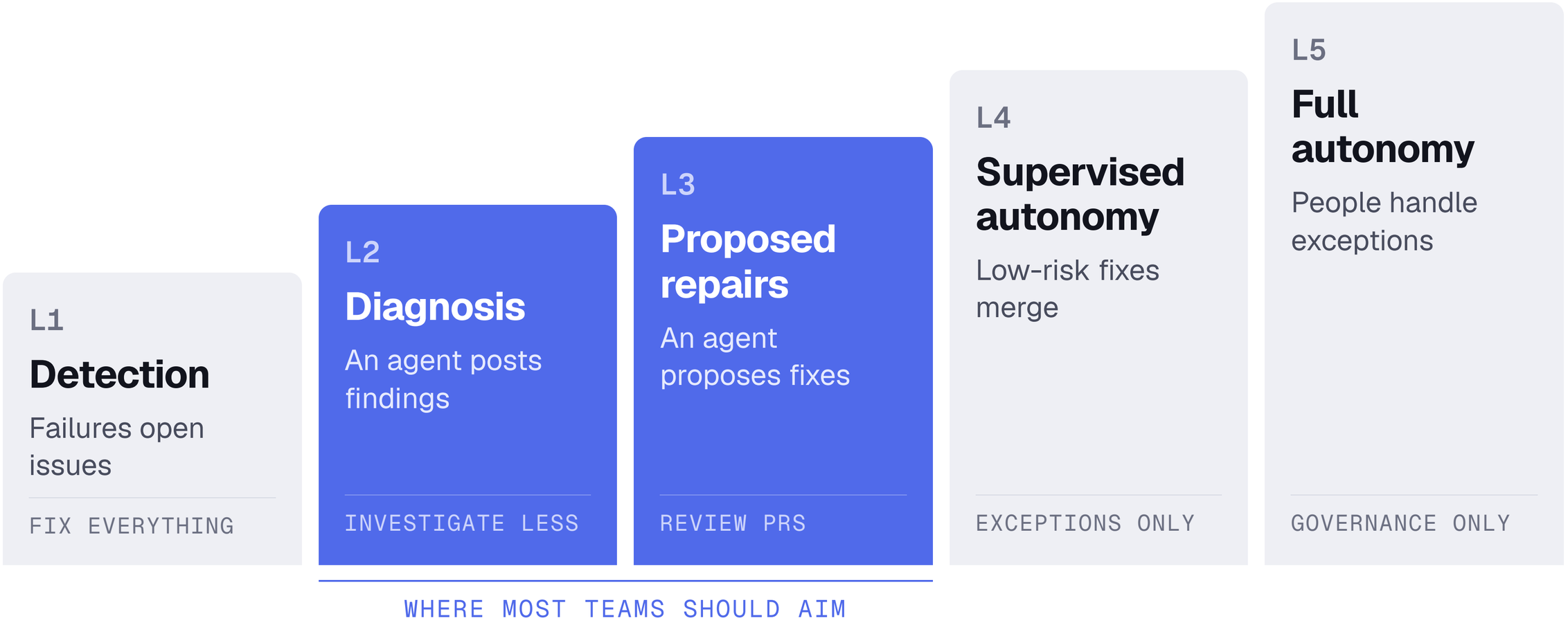
Human judgment rises as the claim gets fuzzier



Relative scale for design purposes, not measured values

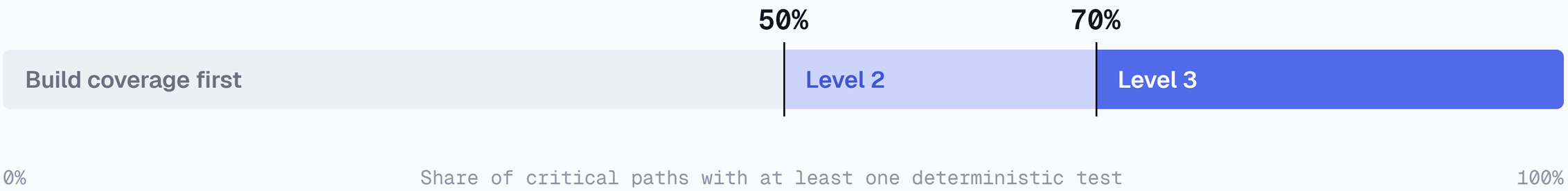
THE MATURITY LADDER

Five levels between "tests file a ticket" and "it merges its own fixes"



BEFORE YOU CLIMB

Deterministic test coverage decides how high you can climb



WHY GATE AT ALL

Below 50%, the diagnosis agent spends its time on false alarms, and the team stops trusting the system.

WHY 70% FOR LEVEL 3

Once an agent proposes changes, you need high confidence that the failures it acts on are real.

WHERE TO STOP

Most teams should target Level 2 or 3



L4 and up need risk tiers, months of proof, and sign-off

SKIP SELF-HEALING WHEN

- Content is regulated
- Coverage is under 50%
- Nobody owns the page
- One author owns a narrow set

BEYOND THE DOCS CUSTOMERS READ

The documentation that breaks isn't always customer-facing



The whole time, internal triage docs called the old behavior "intended"

Lovable, "Our response to the April 2026 incident," April 22, 2026 · cited in *Docs as Tests & AI*, ch. 17

ONE MORE DOC SET

Your agents read docs too.

AGENTS.md files, skill libraries, and agent definitions make claims. They drift like everything else, and they don't warn you when they do.

CONTEXT ROT

When agent instructions go stale, nothing errors

THE API

Deprecates `createUser`

AGENTS.MD

Still says `createUser`

EVERY AGENT

Believes it

YOUR USERS

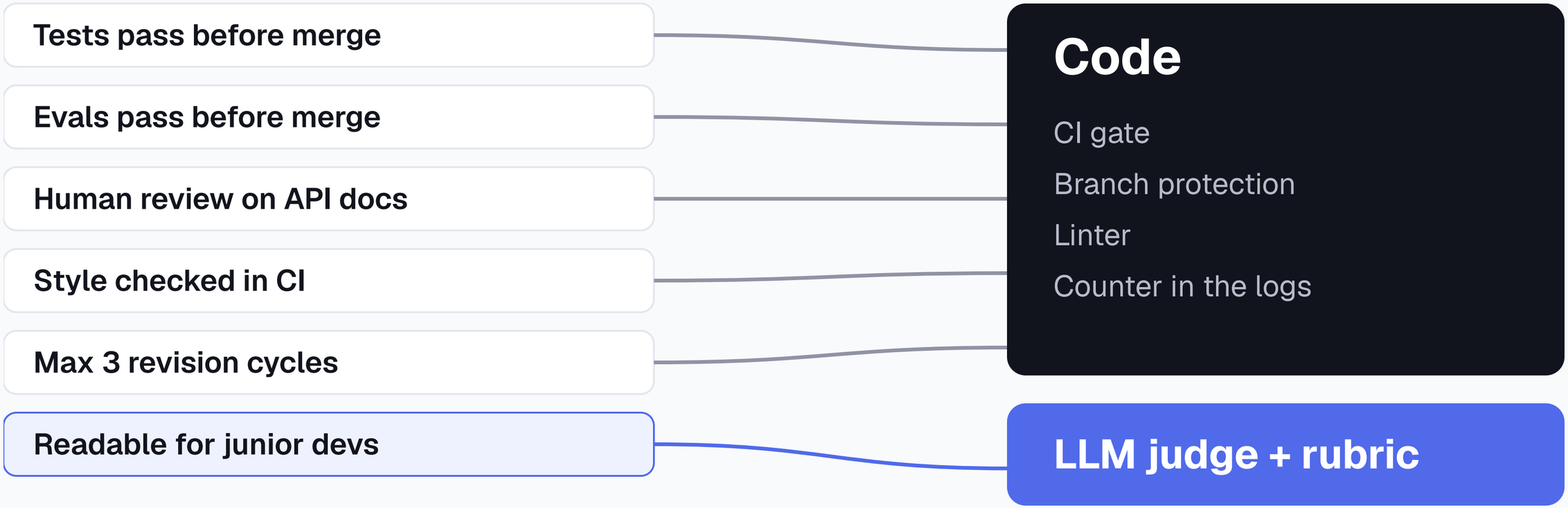
Get code for a dead endpoint

0 errors along the way

Days to trace it back

PROJECT DESCRIPTIONS

Every rule in a project description is already a test



FROM DOCUMENTED TO ENFORCED

Each instruction rule climbs its own ladder

Example rule: "Maximum 3 AI revision cycles before human takeover"

1	**Undocumented**	Someone remembers
2	**Documented**	Written down, unchecked
3	**Monitored**	A weekly report flags it
4	**Enforced**	Pauses at three, pings a person

Only stage 4 catches it when nobody is watching.

SKILL LIBRARIES

An ungoverned skill library wastes context and picks the wrong skill

1,200+
skills in one public catalog

58.5%
with no trigger guidance

13
groups of near-duplicates

47K
tokens before any skill runs

GOVERN IT LIKE A DOC SET

- Says when to use it
- Duplicate check
- An owner
- Evals on model change

Dachary Carey, AEShift, 2026, as cited in *Docs as Tests & AI*, ch. 14

AGENT DEFINITIONS

Tool access is a claim you can test

OVERPRIVILEGED EDITOR

```
tools: Read, Write, Grep,
Glob, Bash, Browser
```

Fixes what it should only flag

SCOPED EDITOR

```
tools: Read, Grep, Glob
```

Produces review comments

WHAT COULD GO WRONG

The pull request looked perfect, and it would have sent 2,000 developers to 404s

Endpoint 1	/v2 → /v3	**Moved**
Endpoint 2	/v2 → /v3	**Didn't**
Endpoint 3	/v2 → /v3	**Didn't**

Over-generalization | Hallucinated fixes | Cascading fixes | Alert fatigue | Masking root causes

Illustrative near-miss and failure modes from *Docs as Tests & AI*, ch. 19

THE SCOREBOARD

Four numbers tell you whether the healing system is healthy

	HEALTHY	WARNING	CRITICAL
Accuracy rate	85%+	80–85%	Under 80%
Rollback rate	Under 3%	3–5%	Over 5%
Over-generalization	Under 5%	5–10%	Over 10%
Fix rate	50%+	30–50%	Under 30%

Starting defaults from *Docs as Tests & AI*, ch. 19. Tighten them for external API reference.

THE SCOREBOARD

Six numbers tell you whether the instructions are healthy

Trigger precision and recall

Fires on the right work

Catalog token overhead

Context spent up front

Duplicate rate

Ambiguous picks

Trigger-guidance coverage

Can it be routed?

Pass rate per version

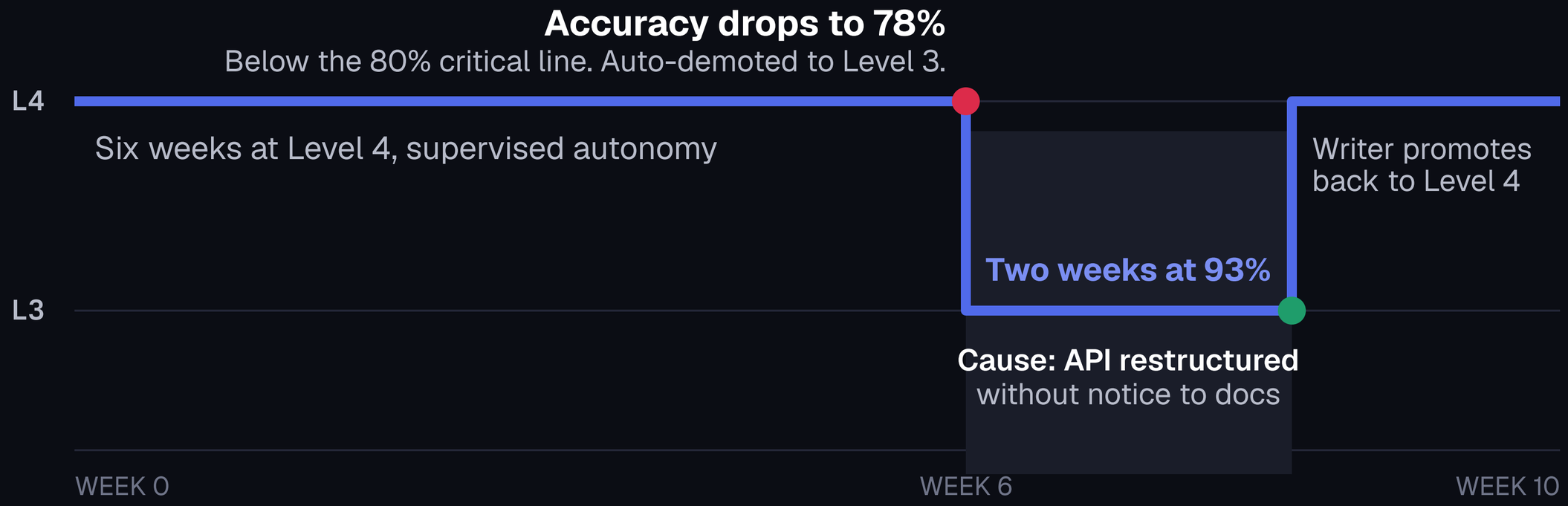
Regressions

Tokens per task

Bloat

A DEMOTION

Demotion is how the system tells you something changed upstream



Payments API reference, illustrative scenario from *Docs as Tests & AI*, ch. 19

WHAT THE STORY TEACHES

Autonomy moves in both directions, one content area at a time

L4 → L3

Demotion keeps autonomy honest

Automatic demotion makes promotion safe

L4 · L3 · L2

Trust is per content area

Each area earns its own level

3 in 30

Repeat healing is a product signal

3 fixes to one page in 30 days? Send a person.

BUILD IT

You can build a basic Level 1 and 2 loop in about a day

TESTING TOOL	CI	ISSUE TRACKER	AI AGENT	PERSON
Runs doc tests	**Runs them on a schedule**	**Opens an issue on failure**	**Investigates**	**Reviews and fixes**
Doc Detective	GitHub Actions	GitHub Issues	GitHub Copilot	Pull request

BUDGET	INTERMEDIATE · AGENT OPENS PRS	ADVANCED · AUTO-MERGE TRIVIAL	PRODUCTION
About 1 day	**1–2 weeks**	**1–2 months**	**Ongoing**

BOUND IT

Put merge permissions in writing before you need them

Trivial

Typos, formatting

After tests pass

Minor

Labels, URLs

Tests + 1-hour delay

Major

Steps added or removed

Manual by default

Breaking

Endpoints, parameters

Never

Product names aren't trivial. Regulated content is always manual.

WHERE THIS STOPS, FOR NOW

Some documentation describes things you can't test on every change

10,643

Cadillac LYRIQs recalled, March 2025

CAMPAIGN	NHTSA 25V-193
DEFECT	The owner's manual inaccurately described the rear outboard head restraints
REMEDY	A corrected manual insert, mailed to every owner

Hardware docs get checked manually, once, and late.

START MONDAY

Four things to do next

Rename your tests for the agent that reads them

What was tested, where, and what it expected.

Wire one failing test to an issue and an agent

A basic Level 2 loop, in about a day.

Treat AGENTS.md and skills as a doc set

Extract their rules. Test the ones a machine can check.

Write thresholds and a never-auto-merge list

Before you climb, while things are calm.

THANK YOU

Questions?

docsastests.com
The strategy, tool-agnostic

doc-detective.com
Open-source doc testing

instructionmanuel.com
My blog, on whatever's on my mind

linkedin.com/in/manuelrbsilva
Let's talk about your setup